



■ CS177 Bioinformatics: Software Development and Applications

Lecture 2: Introduction to Bioinformatics

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Learning Objectives



- At the end of this lecture, you should be able to:
 - Define what is bioinformatics
 - Explain the scope of bioinformatics
 - Tell what skill sets a bioinformatician should have





Basics of Bioinformatics





What is bioinformatics?



- In the first time the term “bioinformatics” was used (in 1970 in a Dutch article), it referred to “the study of informatics processes in biotic systems.”
- Definition by the U.S. National Institutes of Health (NIH) (in 2000):
 “Bioinformatics is the research, development, or application of computational tools and approaches for expanding the use of biological, medical, behavioral or health data, including those to acquire, store, organize, archive, analyze, or visualize such data.”
- In short, “bioinformatics” is the application of information science and technology to the field of molecular biology.

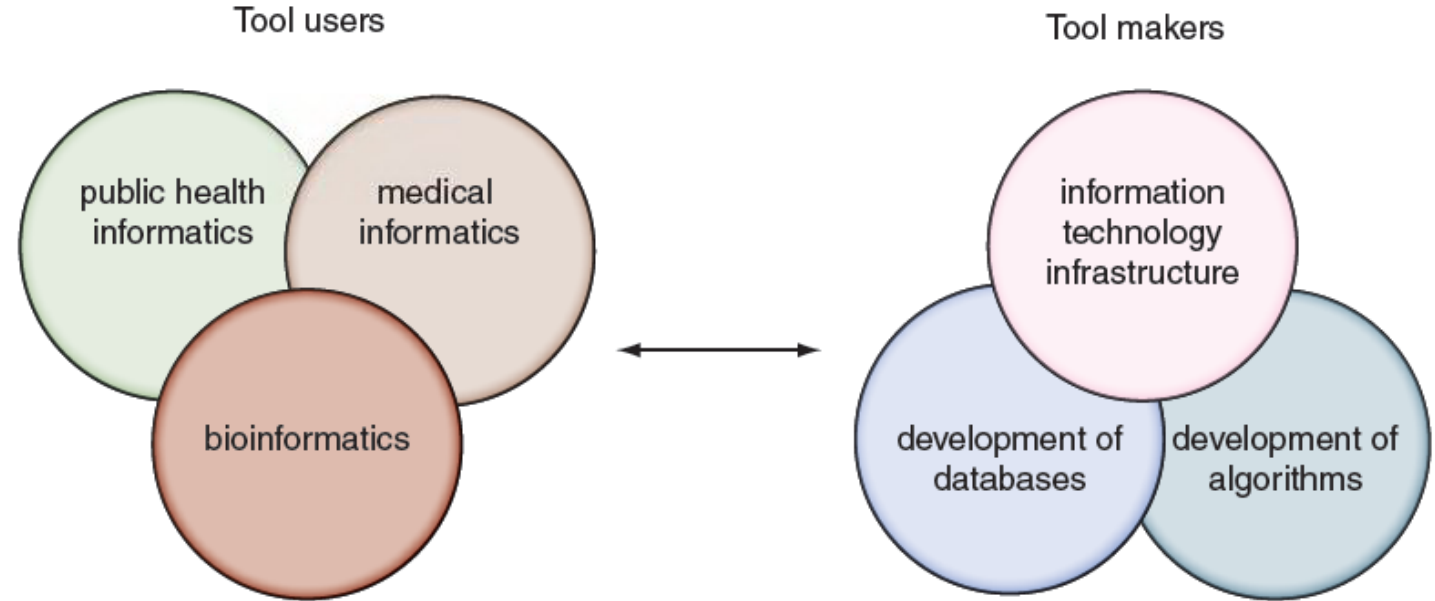




Related fields



- Computational biology
- Computational sciences
- Biostatistics
- Data science
- Theoretical biology
- Mathematical biology
- Medical informatics
- Biological computing



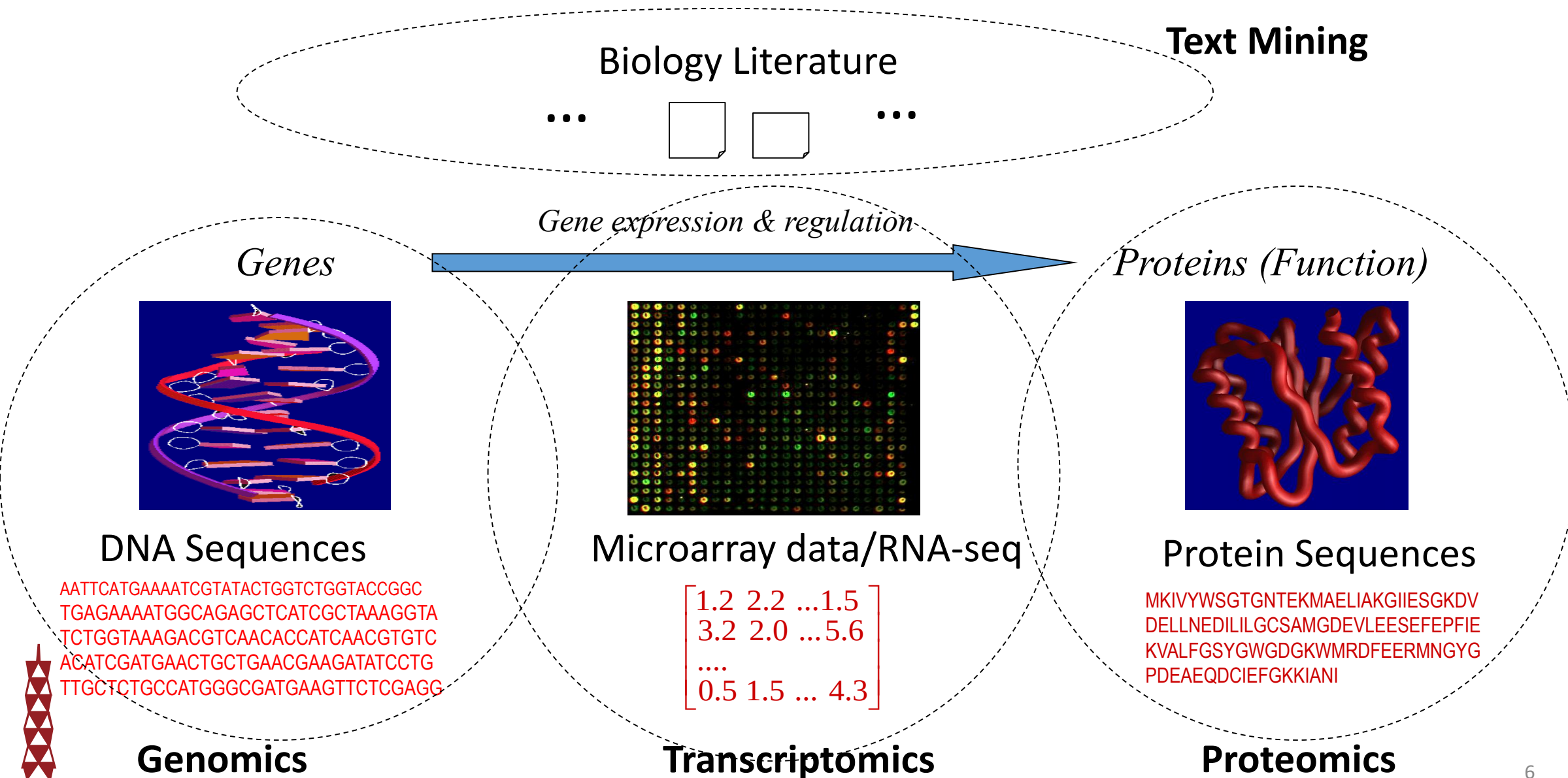
From Pevsner book (Ed. 3, 2015), Fig. 1.6, page 15

The relation between “computational biology” and “bioinformatics”:

- They are often used interchangeably, but there are subtle differences.
- The NIH definition of “**computational biology**” (in 2000): The development and application of data-analytical and theoretical methods, mathematical modeling and computational simulation techniques to the study of biological, behavioral, and social systems.
- Personal view: “computational biology” is focused on “biology” and science, whereas “bioinformatics” is focused on “informatics” and technology. But the two fields have big overlap.



■ Topics in Bioinformatics

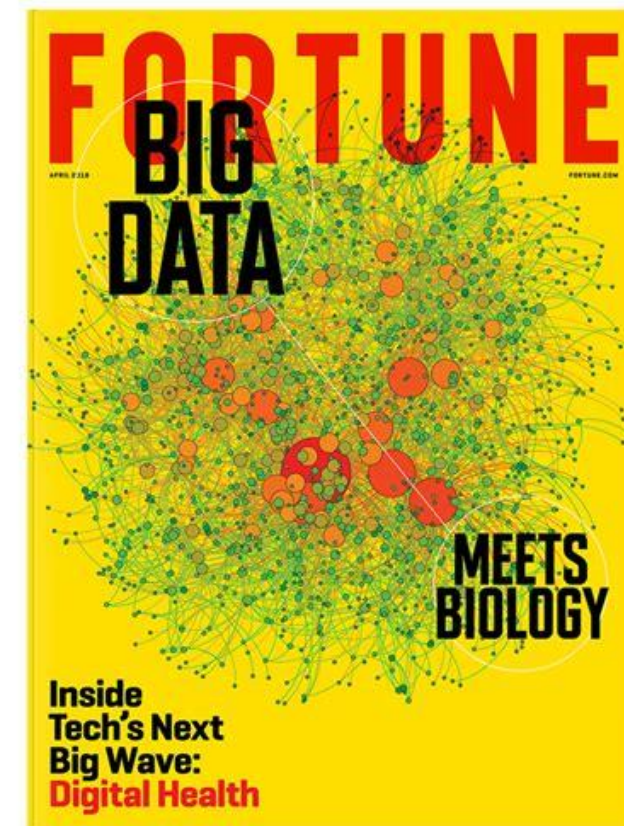




Why is bioinformatics important?



- Lots of biological data are being generated. The **size** and **complexity** of the data make them impossible to analyze without using computers.
- Biological data (especially genetic or genomic sequences) tend to be **discrete** in nature, and thus are ideal for analysis using computers.
- Bioinformatics plays crucial roles in helping biologists and medical researchers to uncover the mysteries of life.
- Bioinformatics could drive further development of biology and biotechnology.



March 19, 2018





A brief history of bioinformatics

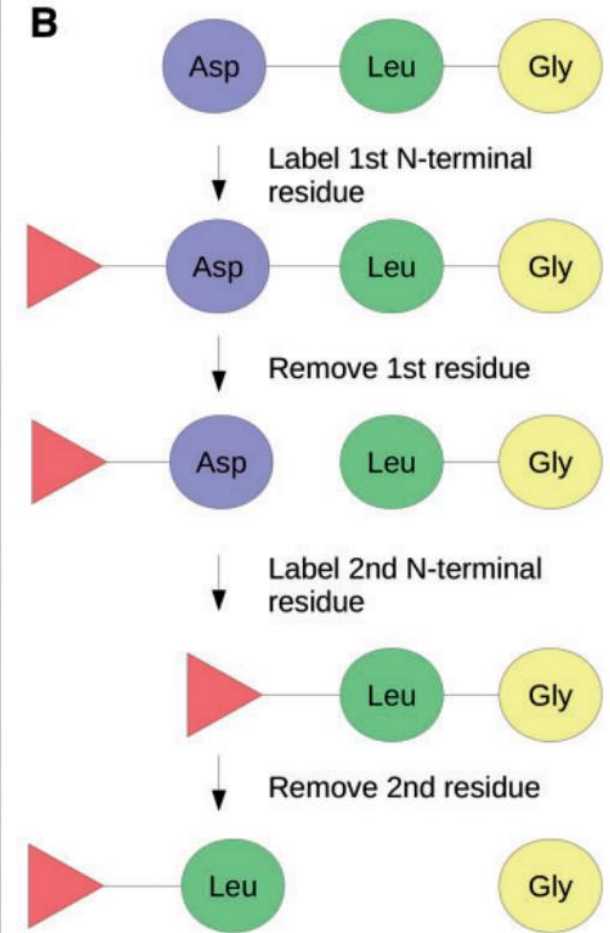


- 1950-1970: The origins
- 1970-1980: Paradigm shift from protein to DNA analysis
- 1980-1990: Parallel advances in biology and computer science
- 1990-2000: Genomics, structural bioinformatics and the information superhighway
- 2000-2010: High-throughput bioinformatics
- 2010-Today: Present and future perspectives



1950s: Start with protein analysis

- 1953: Discovery of double-helix structure of DNA, but sequence unknown as yet
- Determination of protein structures by crystallography
- Protein sequencing:
 - F. Sanger sequenced insulin
 - Automated Edman peptide sequencing
 - **Issue**: Maximum of 50 – 60 amino acids can be sequenced in each reaction
 - **Challenge**: How to assemble a whole protein sequence from hundreds of small sequences?





Dayhoff: the first bioinformatician



- In 1951: Received PhD in chemistry from Columbia University, applied IBM punched-card machines to calculate resonance energies of molecules
- From 1960: as Associate Director of National Biomedical Resource Foundation (NBRF), started collaboration with Robert S. Ledley (a pioneer of biomedical computation)
- 1958–1962: They developed COMPROTEIN software (shown in next slide)
- Developed one-letter amino acid code
- From 1965 (with co-workers): Compiled the first ever biological database (*Atlas of Protein Sequence and Structure*)
- In 1966 (with Richard Eck): first reconstruction of phylogeny by computers from sequence alignment
- In 1978 (with co-workers): Developed the first probabilistic model of amino acid substitutions (PAMs)



Margaret Belle (Oakley) Dayhoff
(1925 – 1983)

American physical chemist
The “mother and father of
bioinformatics”

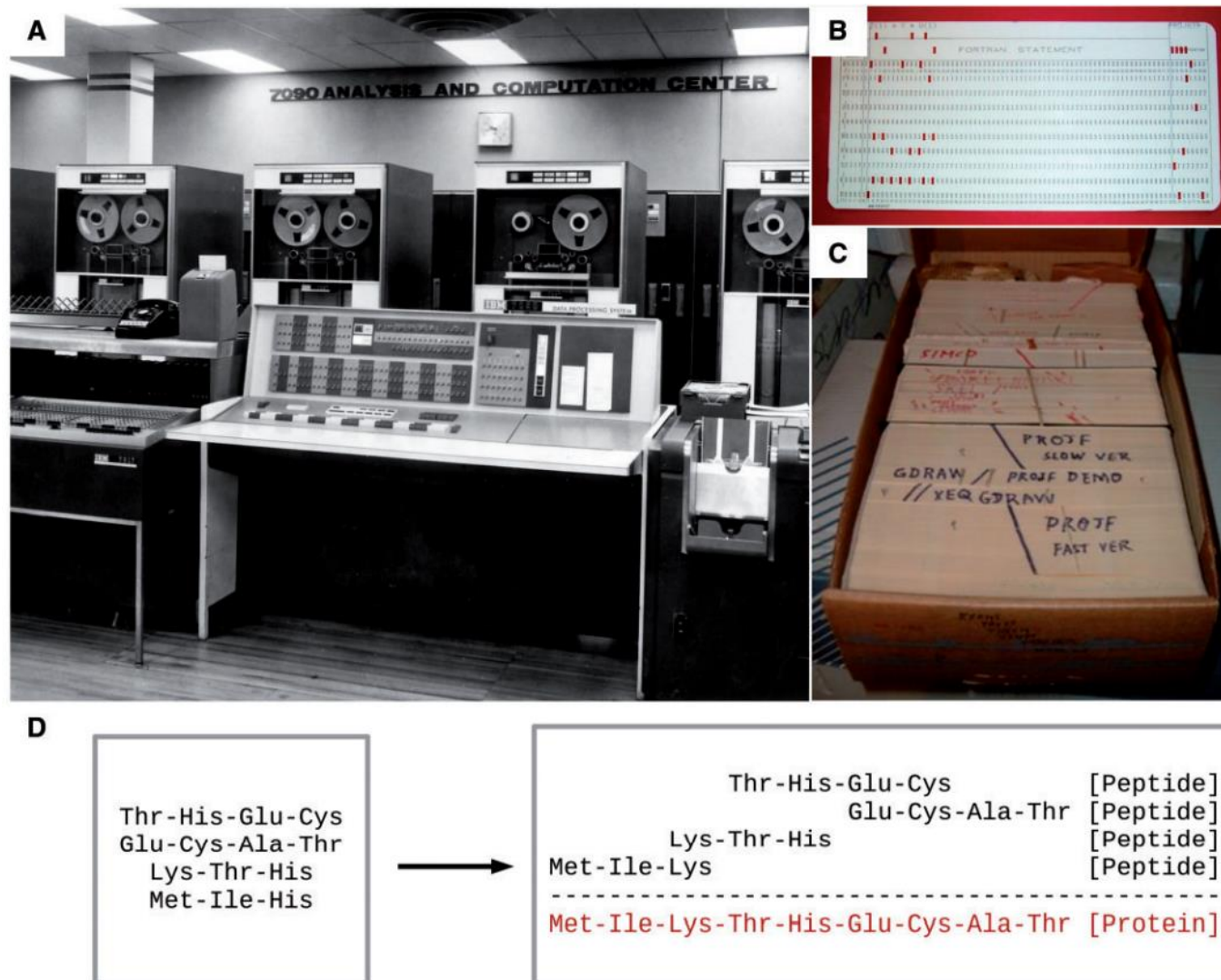


COMPROTEIN: the first bioinformatics software



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- A complete computer program for the IBM 7090 designed to determine protein primary structures using Edman peptide sequencing data – the first *de novo* sequence assembler
- Coded in FORTRAN on punch-cards
- **Input:** Short amino acid sequences represented in 3-letter abbreviations (e.g. Lys for lysine, Ser for serine)
- **Output:** A consensus protein sequence (**red sequence** in Fig. D)





■ 1970-1980: Paradigm shift to DNA analysis

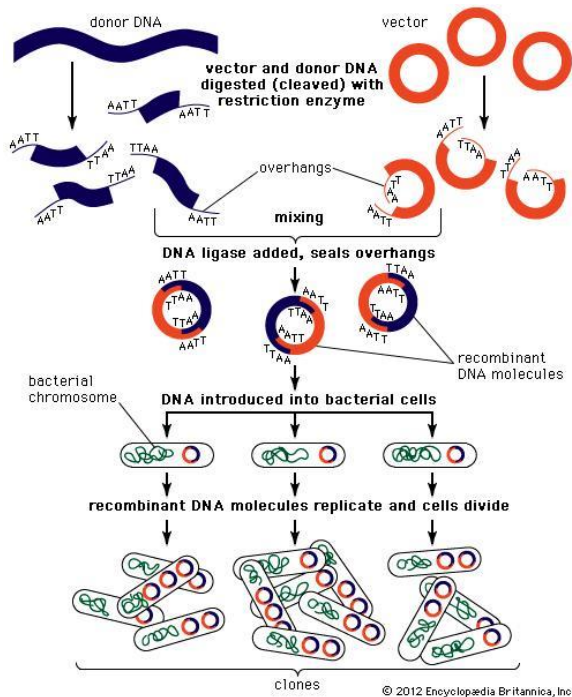
- Deciphering of the **genetic code** (1968):
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 - One can predict the structure of any protein by reading its DNA
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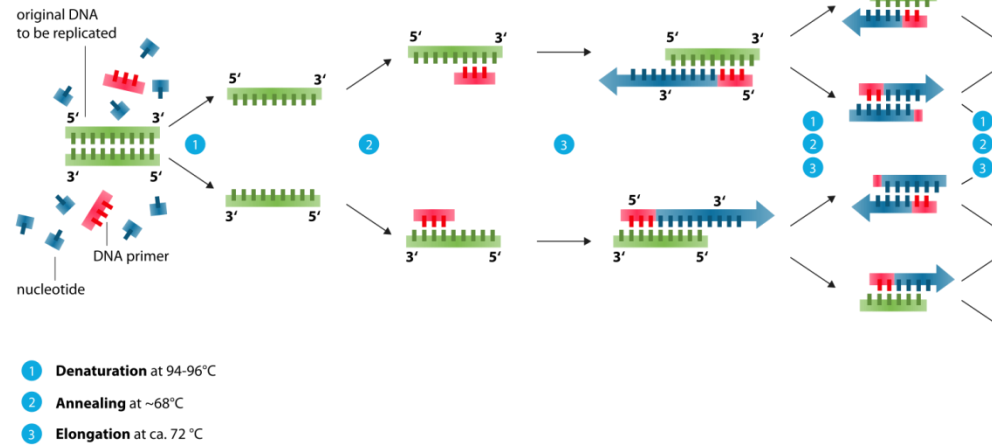
1980–1990: Parallel advances in biology and computer science

Biology

- Molecular methods to target and amplify specific genes
 - Recombinant DNA (1972)
 - Polymerase chain reaction (PCR) (K. Mullis, 1987)
 - Outcome: More sequence data available



Polymerase chain reaction - PCR



Computer

- Access to computers and specialized software
- The free software movement
- Desktop computers
- New programming languages

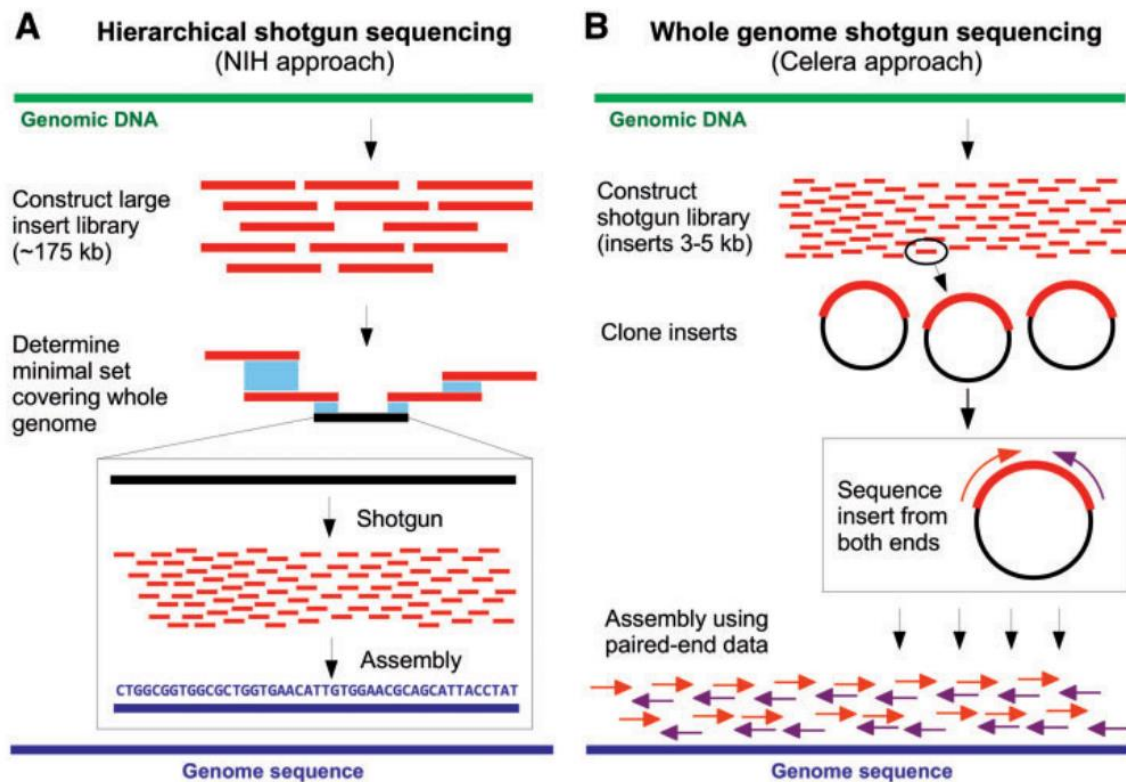


1990-2000: Genomics, structural bioinformatics and the information superhighway



• The Human Genome Project (1990 – 2003)

- Rival between U.S. National Institutes of Health (NIH) and Celera Genomics (led by J. Craig Venter)
- The NIH project costed \$2.7 billion (in 1991 dollars) and over 13 years. By contrast, the cost was \$1000 and less than a week in 2018
- Perl-based software developed in mid to late 1990s to assemble whole-genome sequencing reads, e.g. PHRAP, Celera Assembler, TIGR assembler, MIRA, EULER



Array of ABI 373 capillary sequencers at NIH (1993)

2000-2010: High-throughput bioinformatics



Technology	Read length (bp)	Reads per run	Time per run	Cost per megabase (US\$)	Accuracy (%)
Roche 454	700	1 million	1 day	10	99.90
Illumina	50–250	<3 billion	1–10 days	~0.10	98
SOLiD	50	~1.4 billion	7–14 days	0.13	99.90
Ion Torrent	200	<5 million	2 hours	1	98
Pacific Biosciences	2900	<75,000	<2 hours	2	99
Sanger	400–900	N/A	<3 hours	2400	99.90

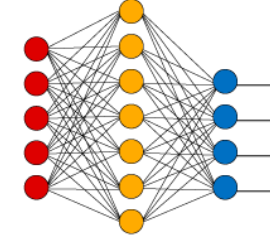
From Pevsner's
book (2015), p. 382

- NGS (next-generation sequencing) technology allowed sequencing millions of DNA molecules in a single run
- Computational problems and applications:
 - Genome assembly
 - Sequence alignment: mapping of millions or billions of short reads to a reference genome
 - Variant calling
 - Visualization of NGS data
- Widely used software for NGS data
 - Newbler (2005): genome assembler
 - Galaxy (2005): workflow system
 - SOAP (2008): short read aligner
 - Bowtie (2009): short read aligner
 - BWA (2009): short read aligner
 - TopHat (2009): RNA-seq data, splicing analysis
 - Cufflinks (2010): RNA-seq data, splicing analysis
 - IGV (2011): NGS data visualization
 - Etc.

2010-2022: Emerging topics

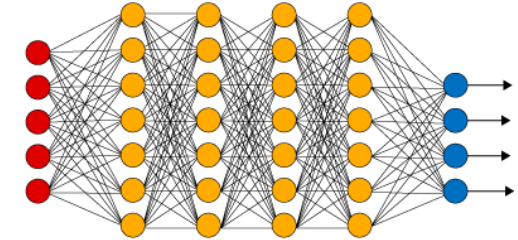
- Epigenomics
- Metagenomics
- Single-cell data analysis
- Personalized medicine and precision medicine
- The rise of artificial intelligence (AI) and data science, especially deep learning
- Systems biology
- Synthetic biology
- Genome editing (CRISPR/cas9, etc.)

Simple Neural Network



● Input Layer

Deep Learning Neural Network

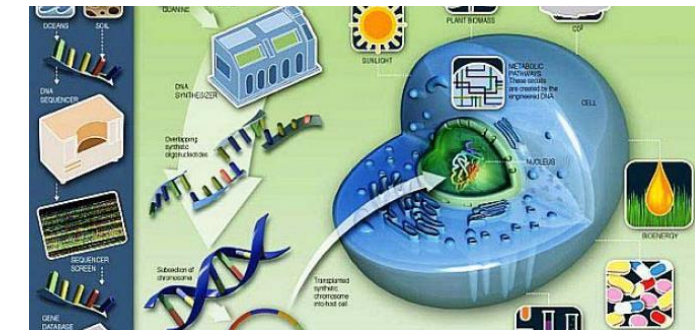


● Hidden Layer

● Output Layer

**'It will change everything':
DeepMind's AI makes
gigantic leap in solving
protein structures**

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

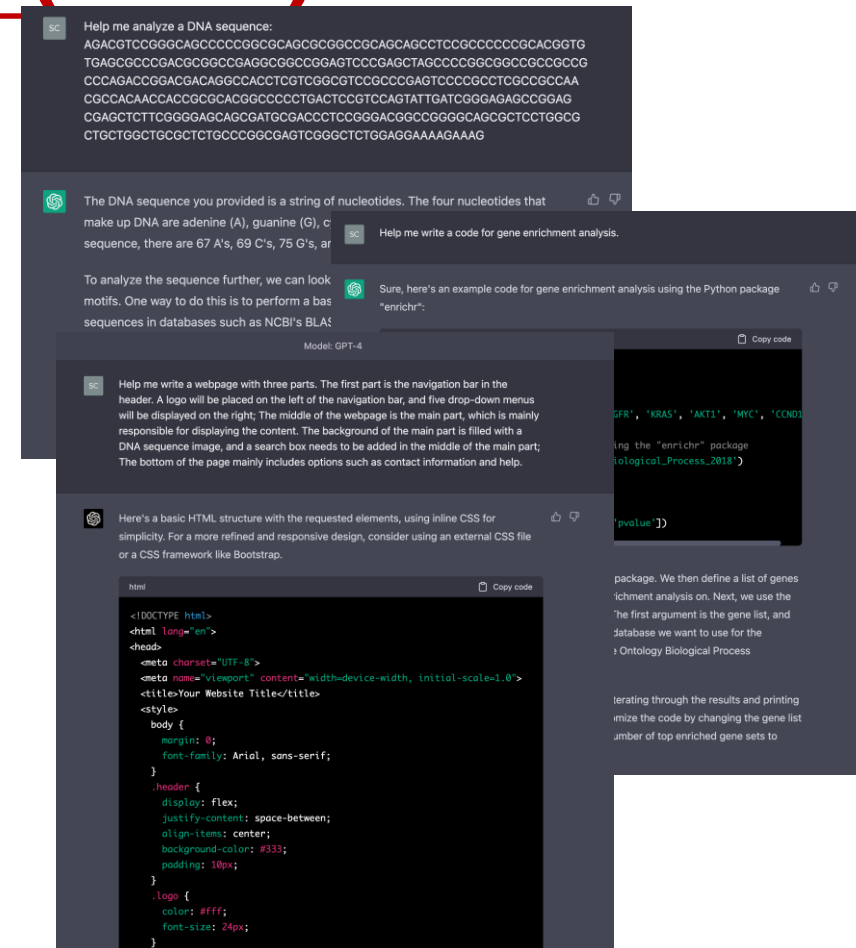
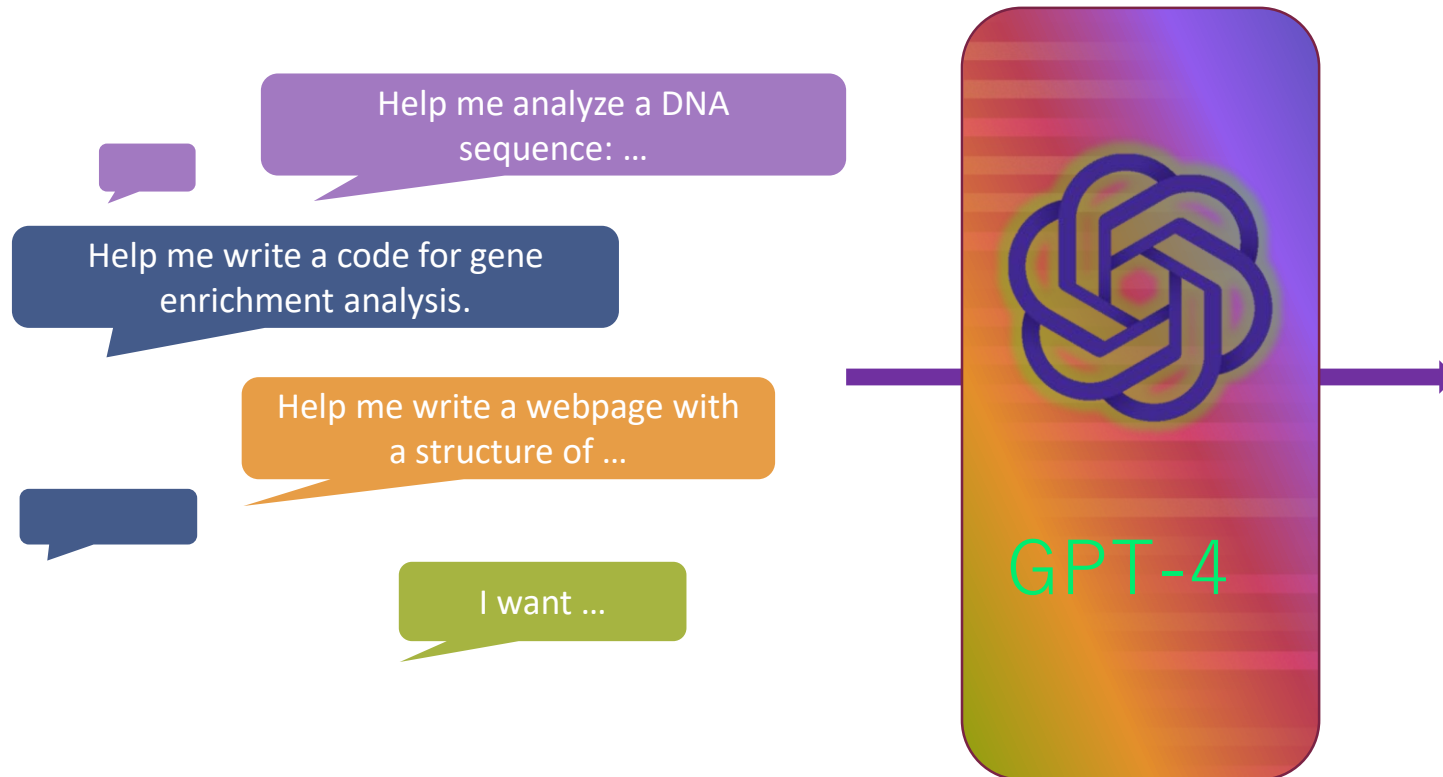


Synthetic biology is the “writing” of DNA, the opposite of DNA sequencing (“reading”)

2023-now: Large language models (LLMs)



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Typical uses of GPT in Bioinformatics:

- Write Python or R code to solve data analysis problems
- Literature survey and paper summarization
- To provide education on biology and computing
- Draft and revise manuscripts
- Recommend software tools and databases, etc.

The dream: No-code Bioinformatics

立志成才 报国裕民



Skill Sets of Bioinformaticians





Three categories of bioinformaticians



- There are basically three categories of professionals involved with bioinformatics:
 - **Bioinformatics users**: access data resources to perform job duties in specific application domains, e.g. experimental biologists, medical professionals (physicians and genetic counselors), legal professionals and biology teachers.
 - **Bioinformatics scientists**: biologists who employ computational methods in order to advance the scientific understanding of living systems.
 - **Bioinformatics engineers**: create the novel computational methods needed by bioinformatics users and scientists.

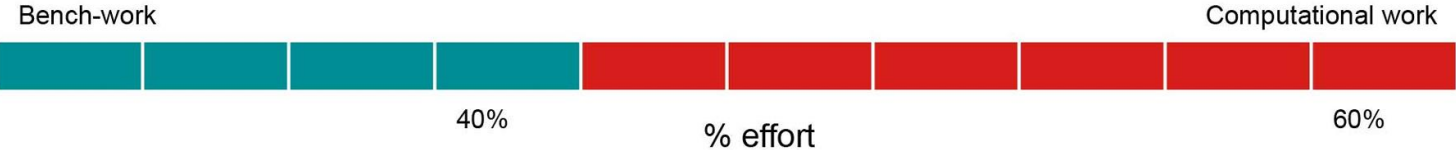


Leon (bioinformatics user)

Leon is on his second postdoctoral fellowship, working on quorum sensing in bacteria. “I’m using a combination of transcriptomics, proteomics and metabolomics to understand these pathogenic changes better” he explains. “I end up with big spreadsheets of protein or gene IDs and I’m trying to piece together which signaling pathways are involved in flipping to the pathogenic state”. He has been on an introductory Unix course but is much more comfortable with GUIs than with the command line. “I just have a visual brain”, he says.



Distribution of time between bench-work and computational work



Preference for using GUI vs command line



Drivers

- Understanding what makes a usually harmless bacterium pathogenic in the lungs of people with cystic fibrosis

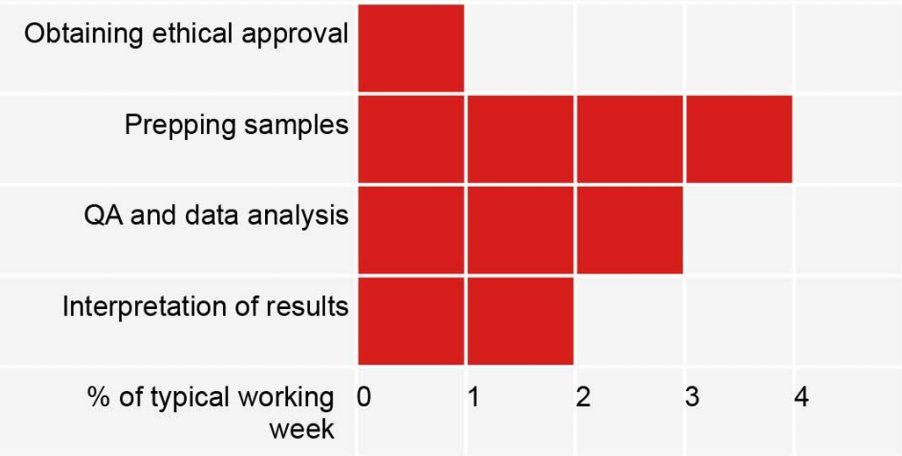
Goals

- QA of -omics data
- Statistical analysis of data
- Data integration and pathway analysis

Pain points

- Lack of access to departmental compute farm
- Sporadic to non-existent access to bioinformatics support

Typical activities



Welch et al. *PLOS Comput. Biol.* 2014



Martha (bioinformatics scientist)

Martha is a senior bioinformatician in an international structural genomics consortium. Her biggest project is on predicting the functions of proteins whose structures have just been solved; she's building a structure-to-function prediction pipeline for the project. This is funded partly by the NIH and partly through industrial funding. She also has a fascination for predicting structure and usually has a student or two working on structural prediction projects.



Distribution of time between bench work and computational work



Preference using for GUI vs command line



Drivers

- Understanding the relationship between sequence, structure and function
- Application to target discovery and validation

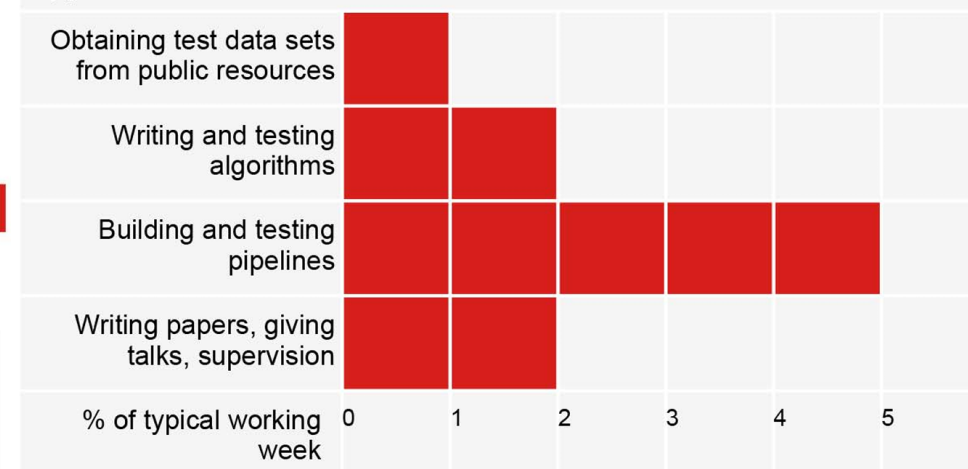
Goals

- Create a structure-to-function pipeline for molecular biologists
- Predict structures de novo from models of similar, solved structures

Pain points

- Sometimes the guys in the lab expect her to fix their computers for them
- Finding students and more senior staff with adequate math

Typical activities



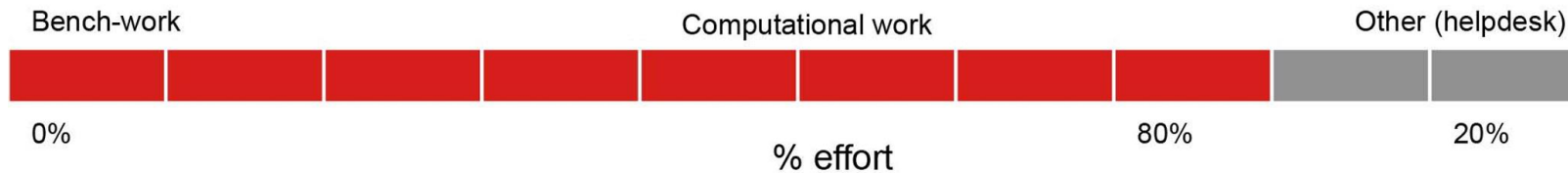
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Ivan (bioinformatics engineer)

Ivan has just started a new support role in a bioinformatics core facility after working for an electronic health records company for four years. His main project is to develop a major new data integration platform for metagenomics data from coral reefs, but he also has to take his share of helpdesk queries on other projects. “I come from a computer science background, so talking the same language as the guys analysing the data is a bit of a challenge,” he says. “I also didn’t really figure that I’d be working on the GUI as well as the code – in my last job we had design folks to take care of that”.



Distribution of time between bench-work and computational work



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Drivers

- Writing algorithms and developing a platform to support novel research
- Supporting other research projects in a busy academic department

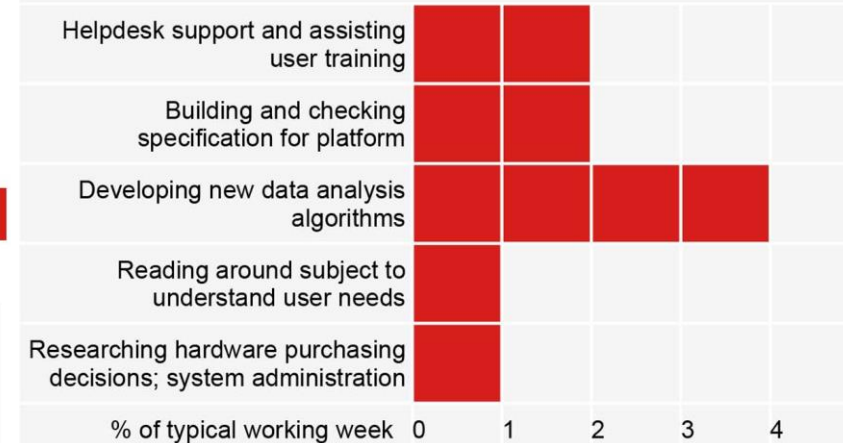
Goals

- Define a spec that meets the needs of his users
- Prototype and build part of the platform
- Make sure his part of the project complements others

Pain points

- Has to work with another software engineer who isn't a team player
- Sometimes struggles to interpret what his users want

Typical activities



Welch et al. *PLOS Comput. Biol.* 2014



Summary of skill sets (before the rise of AI)



Table 1. Summary of the skill sets of a bioinformatician, identified by surveying bioinformatics core facility directors and examining bioinformatics career opportunities.

Skill Category	Specific Skills
General	time management, project management, management of multiple projects, independence, curiosity, self-motivation, ability to synthesize information, ability to complete projects, leadership, critical thinking, dedication, ability to communicate scientific concepts, analytical reasoning, scientific creativity, collaborative ability
Computational	programming, software engineering, system administration, algorithm design and analysis, machine learning, data mining, database design and management, scripting languages, ability to use scientific and statistical analysis software packages, open source software repositories, distributed and high-performance computing, networking, web authoring tools, web-based user interface implementation technologies, version control tools
Biology	molecular biology, genomics, genetics, cell biology, biochemistry, evolutionary theory, regulatory genomics, systems biology, next generation sequencing, proteomics/mass spectrometry, specialized knowledge in one or more domains
Statistics and Mathematics	application of statistics in the contexts of molecular biology and genomics, mastery of relevant statistical and mathematical modeling methods (including experimental design, descriptive and inferential statistics, probability theory, differential equations and parameter estimation, graph theory, epidemiological data analysis, analysis of next generation sequencing data using R and Bioconductor)
Bioinformatics	analysis of biological data; working in a production environment managing scientific data; modeling and warehousing of biological data; using and building ontologies; retrieving and manipulating data from public repositories; ability to manage, interpret, and analyze large data sets; broad knowledge of bioinformatics analysis methodologies; familiarity with functional genetic and genomic data; expertise in common bioinformatics software packages, tools, and algorithms



Core competencies for bioinformatics training (before the rise of AI)



	Bioinformatics User	Bioinformatics Scientist	Bioinformatics Engineer
(a) An ability to apply knowledge of computing, biology, statistics, and mathematics appropriate to the discipline.		X	X
(b) An ability to analyze a problem and identify and define the computing requirements appropriate to its solution.		X	X
(c) An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs in scientific environments.			X
(d) An ability to use current techniques, skills, and tools necessary for computational biology practice.	X	X	X
(e) An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices.			X
(f) An ability to apply design and development principles in the construction of software systems of varying complexity.			X
(g) An ability to function effectively on teams to accomplish a common goal.	X	X	X
(h) An understanding of professional, ethical, legal, security, and social issues and responsibilities.	X	X	X
(i) An ability to communicate effectively with a range of audiences.	X	X	X

Note: Competencies (g) – (n) are common for the 3 categories

Welch et al. *PLOS Comput. Biol.* 2014



References



- Gauthier, et al. "A brief history of bioinformatics" , *Briefings in Bioinformatics*, p. 1-16, 2018.
- Welch, et al. "Bioinformatics curriculum guidelines: toward a definition of core competencies" , *PLOS Computational Biology* 10(3): e1003496, 2014.
- Vincent and Charette. "Who qualifies to be a bioinformatician?" *Frontiers in Genetics*, Vol. 6, Article 164, 2015.





Recap



- What is bioinformatics?
- What can we learn from the history of bioinformatics?
 - The technologies for sequencing of proteins, RNA, and DNA are driving the fields of molecular biology and bioinformatics.
 - The emergence of a new data gathering technology calls for new algorithms and software tools.
 - Advances in information technology provide important driving force.
- What are the skill sets that bioinformaticians need?





Appendix: A Slightly More Complete History of Bioinformaticians





A brief history of bioinformatics

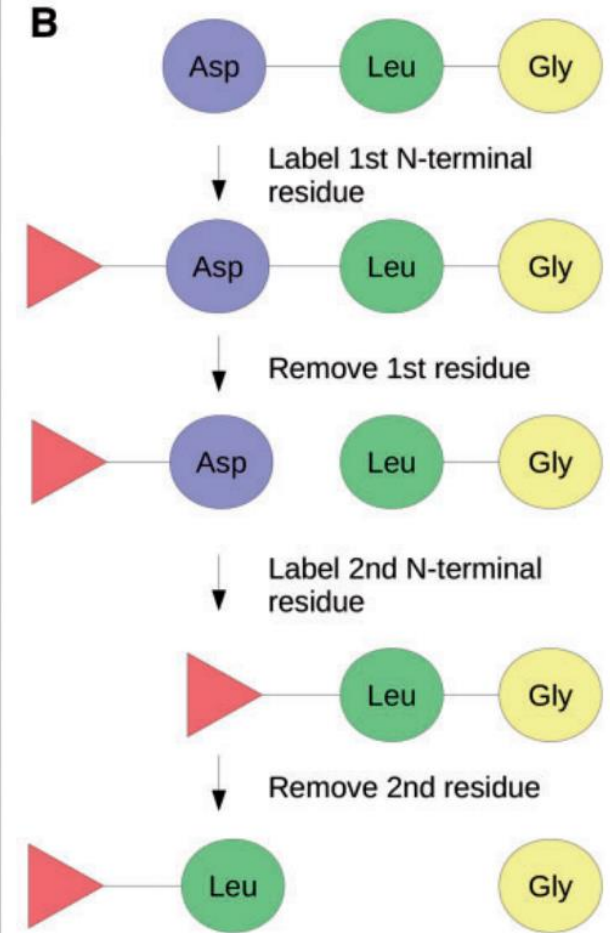


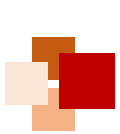
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- 1958–1962: They developed COMPROTEIN software (shown in next slide)
- Developed one-letter amino acid code
- From 1965 (with co-workers): Compiled the first ever biological database (*Atlas of Protein Sequence and Structure*)
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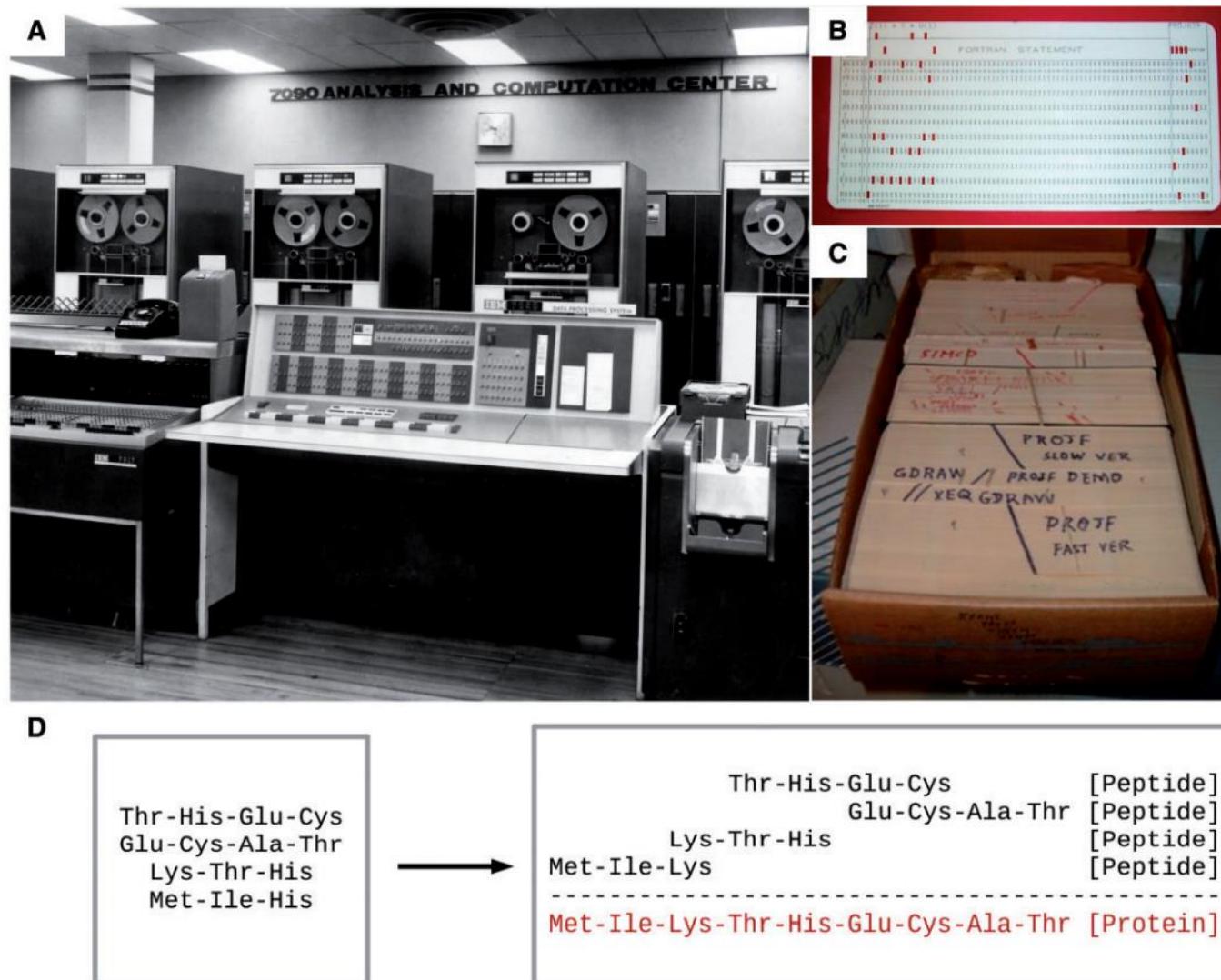


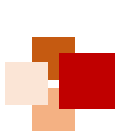
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- **Output:** A consensus protein sequence (**red sequence** in Fig. D)

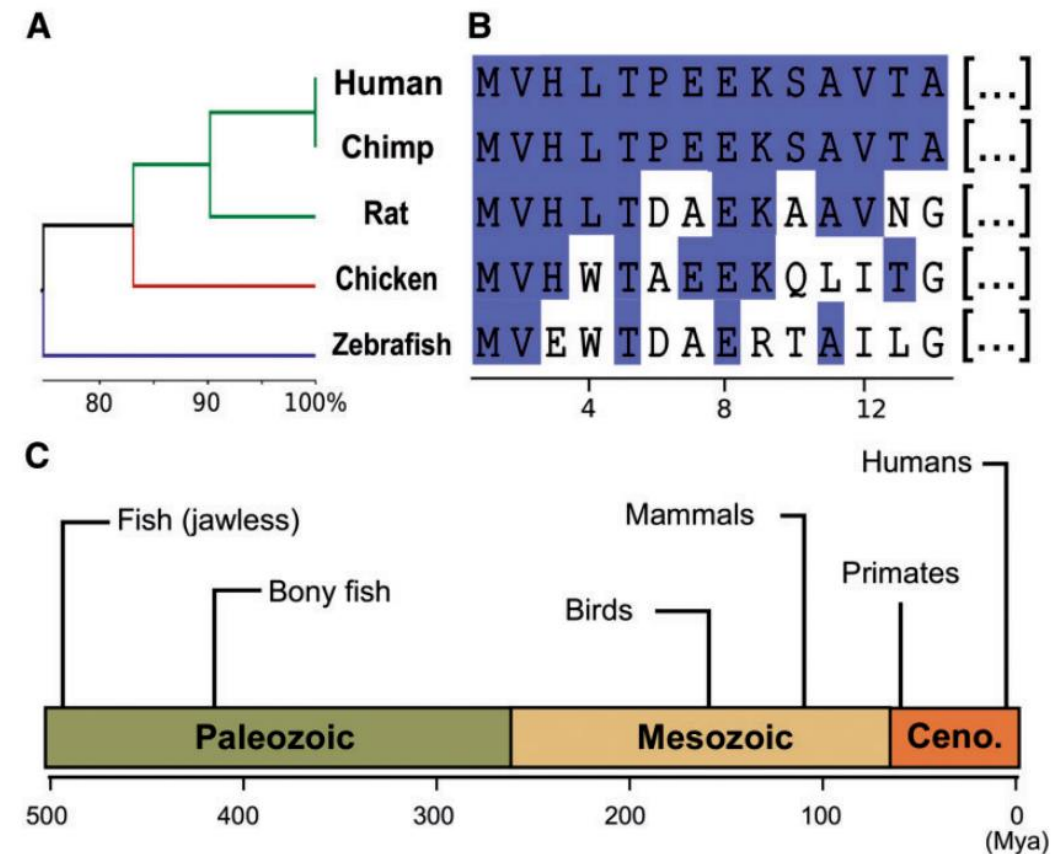




Computer-assisted genealogy of life



- L. Pauling and E. Zuckerkandl (1963): Seminal ideas of molecular evolution, **orthologous** proteins evolved through divergence from a common ancestor
- **Computational challenge**: Design reproducible algorithms for aligning protein sequences
- **Needleman-Wunsch algorithm** (1970): The first **dynamic programming** algorithm for **pairwise protein sequence alignment**
- D-F Feng and R. F. Doolittle (1987): The first practical algorithm for **multiple sequence alignment** (MSA)
 - Progressive sequence alignment
 - (1) perform Needleman-Wunsch alignment for all sequence pairs
 - (2) Use pairwise similarity scores to build a guide tree
 - (3) align the two most similar sequences, and then the next most similar sequence, and so on, according to the guide tree
 - Implemented as **CLUSTAL** software (1988)

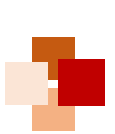




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Using DNA sequences in phylogenetic inference



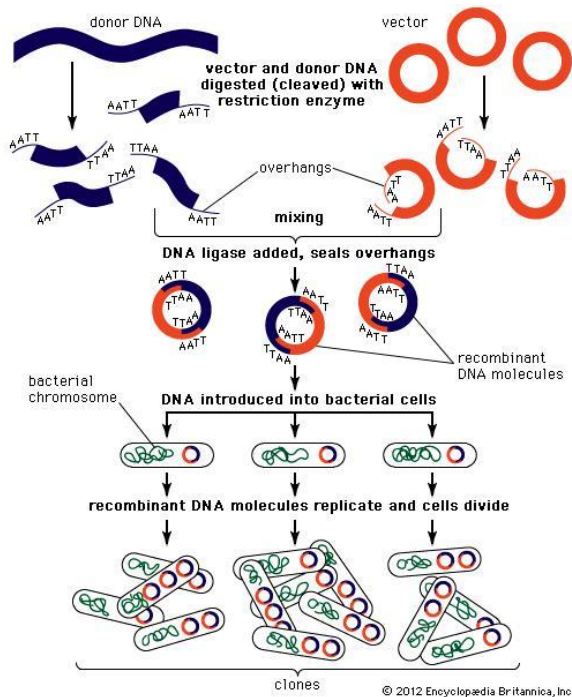
- **Maximum parsimony** method: Reconstruct an evolutionary tree using **the least number of changes**:
 - Build the first molecular phylogenetic trees from protein sequences
 - **Issue**: When data involve moderate to large amounts of changes, the parsimony methods may fail
- Additional information from DNA sequences, compared with protein sequences, e.g.
 - Synonymous mutations (a DNA substitution that does not modify the amino acids due to the degeneracy of the genetic code)
- **Maximum likelihood** method: Find the evolutionary tree which yields **the highest probability of evolving the observed data**:
 - Proposed by Felsenstein in 1981
 - Inspired the use of **Bayesian statistics** in molecular phylogeny in 1990s



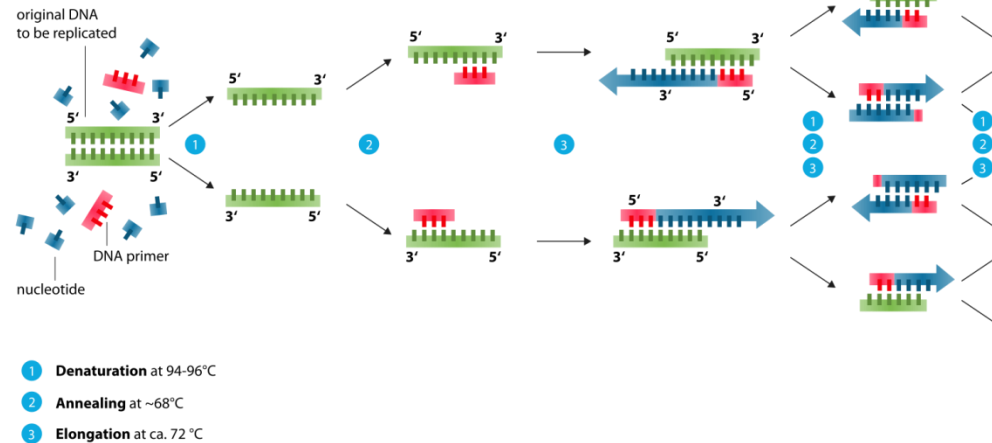
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Polymerase chain reaction - PCR



Computer

- Access to computers and specialized software
- The free software movement
- Desktop computers
- New programming languages



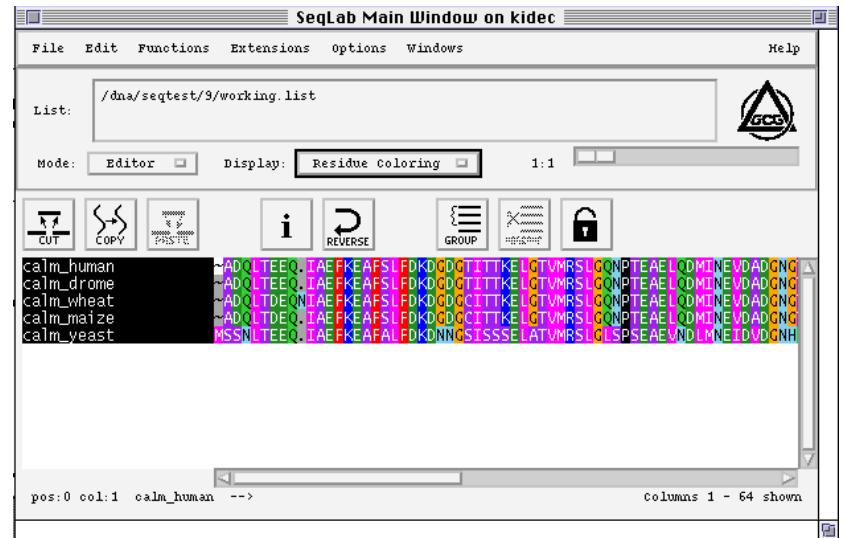
Access to computers and specialized software



- From minicomputer to microcomputers in 1970s
 - Size and weight of minicomputers are cumbersome
 - Lacked user-friendliness
 - Microcomputers appeared in 1977, with built-in BASIC interpreter
- **GCG** software package
 - Published by University of Wisconsin Genetics Computer Group (GCG) in 1984
 - Initially a collection of 33 command-line tools to manipulate DNA, RNA or protein sequences
 - Commercialized in 1990s
- **DNASTAR** software suite
 - Developed soon after GCG in 1984, popular in the 1980s and 1990s for assembling and analyzing Sanger sequencing data
- In the years 1984-1985, other sequence manipulation software suites were developed. Some developers offered free code copies on demand



The DEC VAX-11/780 Minicomputer. The GCG package was initially designed to run on this computer.

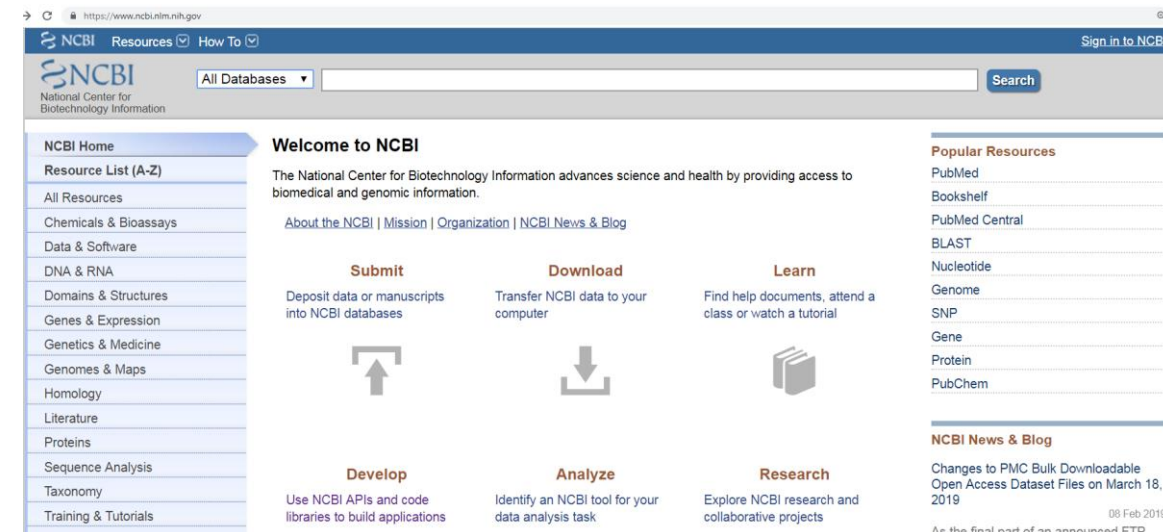


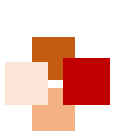


The free software movement



- Richard Stallman created GNU system (in 1985), which grew later as the **Free Software Foundation**, with the philosophy that *"users have the freedom to run, copy, distribute, study, change and improve the software"*
- Initiatives in bioinformatics free software:
 - **EMBOSS** (European Molecular Biology Open Software Suite), later in 1996 as a free and open source alternative to GCG
 - **CCP4** (Collaborative Computational Project Number 4), for macromolecular X-ray crystallography (initiated in 1979)
- Shared databases
 - Sequence data from **EMBL-EBI**, **GenBank** and **DNA Data Bank of Japan (DDBJ)** united (1986-1987)
 - Standardize data formatting, data sharing among the data hubs
 - **NCBI (National Center for Biotechnology Information)** was founded in 1988, hosting databases and software (e.g. GenBank, PubMed, BLAST)
- Dedicated scientific journals:
 - **CABIOS** (*Computer Applications in the Bioscience*) was established in 1985, now named *Bioinformatics*





Desktop computers and new programming languages

- Desktop computers available:
 - Advent of x86 and RISC microprocessors in early 1980s
 - Desktop workstations run on Unix operating systems
- Programming Languages:
 - Non-scripting: Fortran, C, Java
 - Scripting: Perl (1987), Python (1989), R (1993)

Table 2. Selected early bioinformatics software written in Perl

Software	Year released	Use	Reference
GeneQuiz	1994 (oldest)	Workbench for protein sequence analysis	[65]
LabBase	1998	Making relational databases of sequence data	[66]
Phred-Phrap-Consed	1998	Genome assembly and finishing	[67]
Swissknife	1999	Parsing of SWISS-PROT data	[68]
MUMmer	1999	Whole genome alignment	[69]

PubMed Key: (perl bioinformatics) AND ("1987"[Date-Publication]:"2000"[Date-Publication]).

Table 3. Notable nonscripting and/or statistical programming languages used in bioinformatics

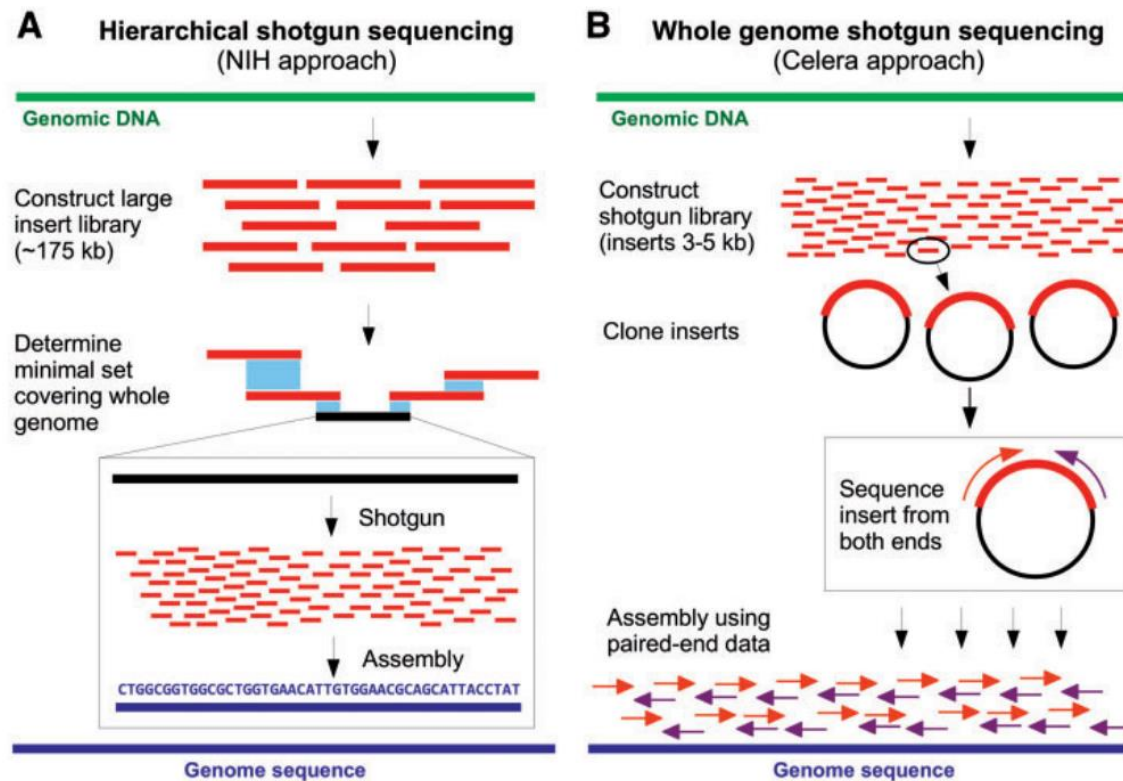
	Fortran ^a	C	R	Java
First appeared	1957	1972	1993	1995
Typical use	Algorithmics, calculations, programming modules for other applications	Optimized command-line tools	Statistical analysis, data visualization	Graphical user interfaces, data visualization, network analysis
Notable fields of application	Biochemistry, Structural Bioinformatics	Various	Metagenomics, Transcriptomics, Systems Biology	Genomics, Proteomics, Systems Biology
Specialized bioinformatics repository?	None	None	Bioconductor, [73], since 2002	BioJava [74], since 2002
Example software or packages	Clustal [32, 33], WHAT IF [75]	MUSCLE [76], PhyloBayes [77]	edgeR [78], phyloseq [79]	Jalview [80], Jemboss [81], Cytoscape [82]



1990-2000: Genomics, structural bioinformatics and the information superhighway

• The Human Genome Project (1990 – 2003)

- Rival between U.S. National Institutes of Health (NIH) and Celera Genomics (led by J. Craig Venter)
- The NIH project costed \$2.7 billion (in 1991 dollars) and over 13 years. By contrast, the cost was \$1000 and less than a week in 2018
- Perl-based software developed in mid to late 1990s to assemble whole-genome sequencing reads, e.g. PHRAP, Celera Assembler, TIGR assembler, MIRA, EULER



Array of ABI 373 capillary sequencers at NIH (1993)

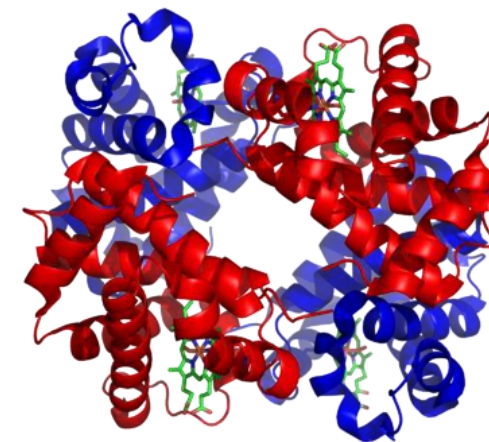
- In early 1990s, Tim Berners-Lee initiated the World Wide Web at CERN
- Many bioinformatics resources became accessible on the web:
 - EMBL Nucleotide Sequence Data Library (the world's first nucleotide sequence database) along with databases like [SWISS-PROT](#) and [REBASE](#) on the web in 1993
 - [GenBank](#) moved to NCBI in 1992
- NCBI website available online in 1994
 - [BLAST](#): allows to perform pairwise sequence alignment efficiently
 - Establishing major databases: [Genomes](#) (1995), [PubMed](#) (1997), [Human Genome](#) (1999)
- Bioinformatics software tools usable through web-servers
 - Traditional software often required Unix-like command lines and the installation of software libraries (dependencies)
 - Web-server software tools have graphical user interface (GUI) and require no installation
 - Journal *Nucleic Acids Research* publishes a [web-server issue](#) each year
- Use internet for scientific publications (i.e. online journals and preprint servers)
 - Cornell University's arXiv (est. 1991)
 - Cold Spring Harbor's bioRxiv (est. 2013)



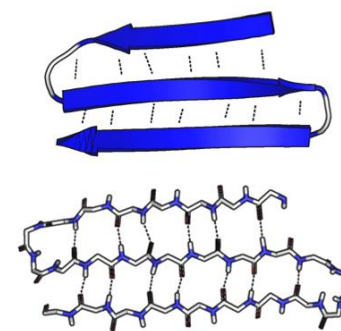
Beyond sequence: structural bioinformatics



- In 1951, Pauling and Corey reported the prediction of α -helices and β -sheets
- In 1958, the first 3D structure of a protein (myoglobin) was experimentally determined using X-ray diffraction
- In 1990s, software tools to model the molecular dynamics (MD) of proteins were released:
 - Based on biophysics breakthrough: force fields were developed to describe the interactions between atoms
 - Require large amounts of computational resources (e.g. in 1998, a microsecond simulation of a 36-amino-acid peptide required months of calculations with 256 CPUs)
- The use of GPUs (graphics processing units) for MD simulations and other bioinformatics applications (e.g. the reconstruction of molecular phylogenies)



The Hemoglobin molecule, made largely of α -helices



β -sheets (3 strands)

2000-2010: High-throughput bioinformatics



Technology	Read length (bp)	Reads per run	Time per run	Cost per megabase (US\$)	Accuracy (%)
Roche 454	700	1 million	1 day	10	99.90
Illumina	50–250	<3 billion	1–10 days	~0.10	98
SOLiD	50	~1.4 billion	7–14 days	0.13	99.90
Ion Torrent	200	<5 million	2 hours	1	98
Pacific Biosciences	2900	<75,000	<2 hours	2	99
Sanger	400–900	N/A	<3 hours	2400	99.90

From Pevsner's
book (2015), p. 382

- NGS (next-generation sequencing) technology allowed sequencing millions of DNA molecules in a single run
- Computational problems and applications:
 - Genome assembly
 - Sequence alignment: mapping of millions or billions of short reads to a reference genome
 - Variant calling
 - Visualization of NGS data
- Widely used software for NGS data
 - Newbler (2005): genome assembler
 - Galaxy (2005): workflow system
 - SOAP (2008): short read aligner
 - Bowtie (2009): short read aligner
 - BWA (2009): short read aligner
 - TopHat (2009): RNA-seq data, splicing analysis
 - Cufflinks (2010): RNA-seq data, splicing analysis
 - IGV (2011): NGS data visualization
 - Etc.



Biological big data

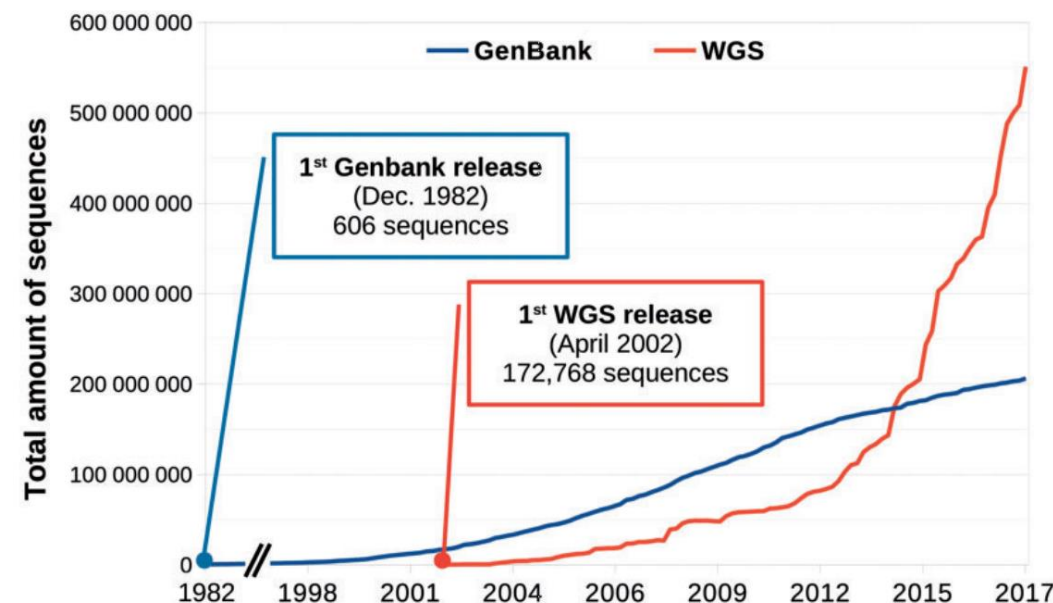


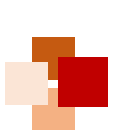
- Big Data challenges

- An exponential increase of sequences in public databases (e.g. GenBank and WGS)
- Major computational resources are needed to handle the data (store, process and organize for access)
- Need new repositories for model species and general genomic databases
- Need to set standards to structure the data resources, e.g. Genomic Standards Consortium created in 2005

- High-performance and collaborative computing:

- GPGPU became popular
- Cloud computing emerged, e.g. Amazon AWS launched in 2006
- Community computing, e.g. BOINC (Berkeley Open Infrastructure for Network Computing) started in 2002





2010-Today: Present and future perspectives



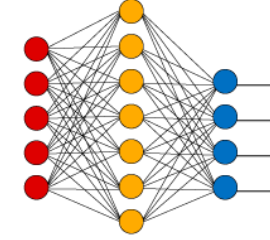
- Recent trends in bioinformatics:
 - Emergence of “bioinformatician” as a profession
 - Increase of user-friendly software tools, often through integrative web servers (e.g. Galaxy)
 - Increase of helping communities, e.g. SEQanswers, BioStar
 - Explosive need for bioinformaticians on the job market in academic, private and government sectors
 - New generations of life scientists are increasingly familiar with computers
 - The booming of AI (esp. deep learning)



2010-Today: Emerging topics

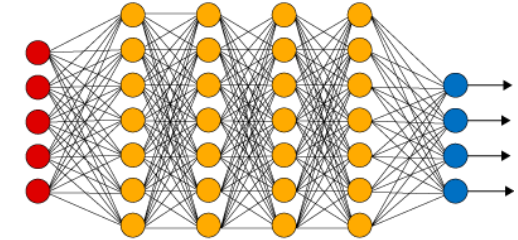
- Epigenomics
- Metagenomics
- Single-cell data analysis
- Personalized medicine and precision medicine
- The rise of artificial intelligence (AI) and data science, especially deep learning
- Systems biology
- Synthetic biology
- Genome editing (CRISPR/cas9, etc.)

Simple Neural Network



● Input Layer

Deep Learning Neural Network

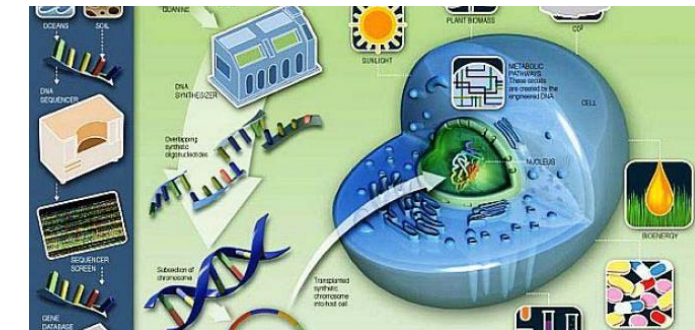


● Hidden Layer

● Output Layer

**'It will change everything':
DeepMind's AI makes
gigantic leap in solving
protein structures**

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.



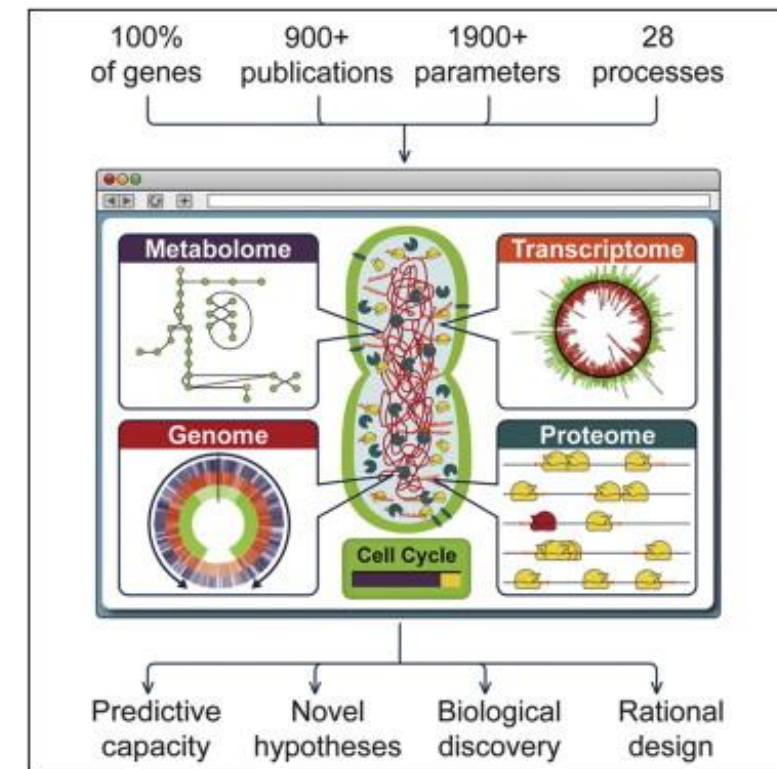
Synthetic biology is the “writing” of DNA, the opposite of DNA sequencing (“reading”)



Systems biology: modeling life as a whole



- **Systems biology** is “the study of the interactions and behaviour of the components of biological entities, including molecules, cells, organs, and organisms” (from Encyclopedia Britannica)
- The philosophy:
 - The whole is greater than the sum of the parts
 - A **holistic** approach to deciphering the complexity of biological systems
 - **Integrate** multiple disciplines (biology, computer science, physics, engineering), different types of data and knowledge
- Whole-cell modeling
 - Single-cell organism of *Mycoplasma genitalium* (in 2012)
 - All its genes, their products and their known metabolic interactions have been reconstructed *in silico*
- Future: To model a whole multi-cell organism



The first Whole-cell model for the human pathogen *Mycoplasma genitalium* (Karr et al. *Cell*, 2012)

